07. FORMATION OF THE OPTIC PLACODE

DURATION : 04:28

STUDY SHEET

COURSE SUMMARY

In this video, we delve deeply into the formation of the optic placode, a key event in embryonic development. From the ectoblast and neural tube, the first optic anlage emerges, revealing the importance of cellular communication in various forms, including autocrine, paracrine, and juxtacrine, and its interaction with genetic phenomena such as sonic hedgehog (HH). We explore the formation of the optic vesicle and the transformation of the epiblast into the primitive optic placode, highlighting the influence of surrounding structures such as the notochord and the colobomic fissure in ocular morphogenesis. This content is essential for any student or therapist wishing to deepen their understanding of the embryological development of the eye.

FORMATION OF THE OPTIC PLACODE

In this section, we will explore the **formation of the optic placode** from the surface ectoblast and the neural tube.

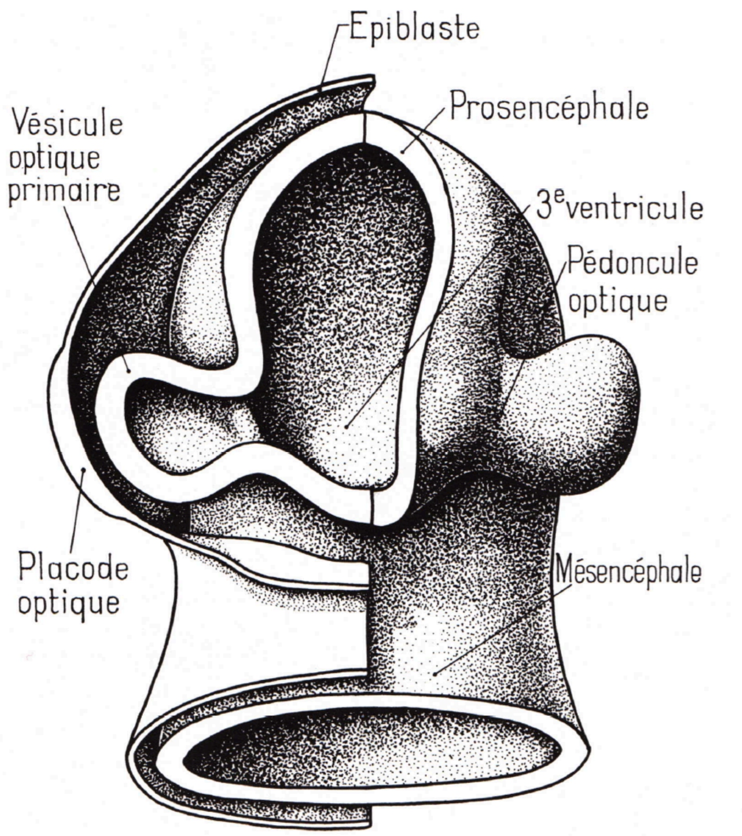

FIG. — Neural tube and forming optical primordia

We first observe the **neural tube** internally, with the **neuropore** and the **cephalic groove** which is not yet closed. It is at this stage that the first **optic anlage** appears. This appearance is due to an **expansion** by points of support outwards.

TYPES OF CELL COMMUNICATION

It is essential to understand the different types of **cell communication**:

1. **Autocrine**: The cell informs itself directly.

2. **Paracrine**: The cell informs its neighbour through direct contact.

3. **Telocrine**: Long-distance communication, as in the case of **endocrine** hormones.

4. **Neurocrine**: Communication via neurotransmitters.

5. **Juxtacrine**: Communication between adjacent cells.

In the case of the optic placode, the information from the ectoblast is linked to a genetic phenomenon, notably **sonic hedgehog (HH)**, which is inhibited at the tip of the prechordal plate. This leads to **lateral expansion** through various genetic phenomena.

FORMATION OF THE OPTIC VESICLE

We observe the **primary optic vesicle** beginning to form. Two lateral vesicles appear and push on the part of the surface epiblast. A **juxtacrine metabolic reaction** occurs, leading to swelling which forms the **primitive optic placode**.

Initially, an epiblastic tissue appears between the **seventh and eighth day**. This tissue evolves into a **neural plate**, which, under the influence of the **notochord**, changes shape to become a **groove**. This groove closes to form a **neural tube**. The upper part disappears, leaving the **neural crest** and the surface epiblastic tissue.

AMNIOTIC FLUID AND CSF

The fluid present above the optic placode is the same as the fluid inside, i.e., the **amniotic fluid**, which will subsequently become the **cerebrospinal fluid (CSF)**.

The small lateral vesicles swell and, thanks to **juxtacrine** molecular contact, the epithelial tissue modifies to form the **optic placode**. This placode becomes a **fulcrum**, giving the impression of sinking in the drawing. This is due to growth from the outside, which creates a point of support.

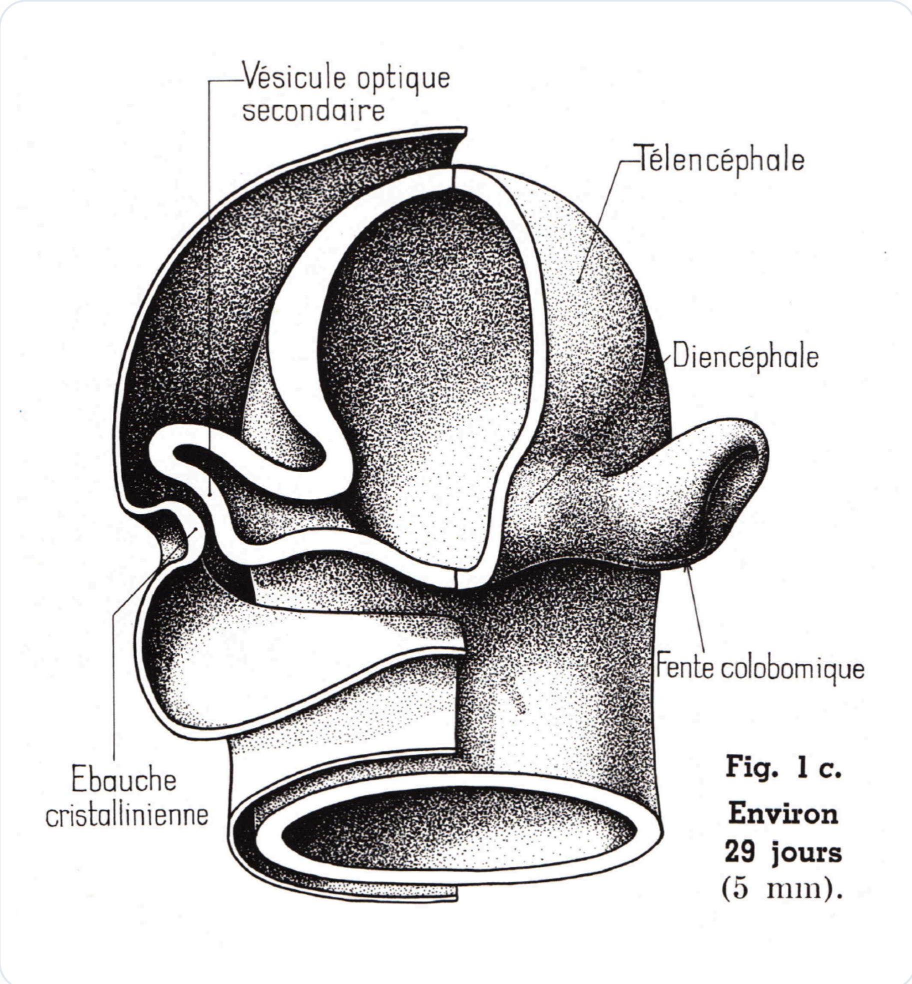

FIG. — Optical placode thickening and acting as a fulcrum

COLOBOMIC CLEFT

It is important to note that there is a **colobomic cleft** through which an artery passes, known as the **hyaloid artery**. This cleft plays a crucial role in the development of the optic placode and surrounding structures.